

# DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: System Software and Compiler Design | Course Code: 21CS61 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

## INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

### MODULE 1

- Q1.** For the grammar:  $E \rightarrow T E_{\text{prime}}$ ,  $E_{\text{prime}} \rightarrow + T E_{\text{prime}} \mid \epsilon$ ,  $T \rightarrow F T_{\text{prime}}$ ,  $T_{\text{prime}} \rightarrow * F T_{\text{prime}} \mid \epsilon$ ,  $F \rightarrow ( E ) \mid \text{id}$ . Compute FIRST and FOLLOW sets for all non-terminals and construct the LL(1) predictive parsing table.

[8 Marks]